

1. Administrative & Study Identification

Full Study Title: An open-label, adaptive design study in patients with amyotrophic lateral sclerosis (ALS) to characterize safety, tolerability and brain microglia response, as measured by TSP0 binding, following multiple doses of BLZ945, using positron emission tomography (PET) with the radioligand [11C]-PBR28 **Short Title/Acronym:** Study of Safety and Proof of the Mechanism of BLZ945 in ALS Patients **Protocol Number:** CBLZ945C12201 (EUDRACT: 2019-000826-22) **NCT Number:** NCT04066244 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** BLZ945 (sotuletinib) **Phase of Development:** Phase II **Medical Monitor:** Novartis Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To characterize the safety, tolerability, and brain microglia response to sotuletinib (BLZ945) as measured by TSP0 binding using positron emission tomography (PET) with the radioligand [11C]-PBR28 following multiple doses. **Secondary Objectives:** To evaluate additional safety and pharmacokinetic parameters, including the mean change from baseline in Left Ventricular Ejection Fraction (LVEF) and the time to reach maximum drug concentration (Tmax).

2.2 Background & Rationale To address the critical unmet medical need in Amyotrophic Lateral Sclerosis (ALS), a fatal neurodegenerative disease. Neuroinflammation, driven by microglial activation, is strongly implicated in ALS progression. This trial was designed to provide proof of mechanism for BLZ945 (sotuletinib), a compound intended to target and modulate these pro-inflammatory microglial responses. *(Note: The sponsor terminated the study after an assessment of the potential benefit-risk profile from available data.)*

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A (Adaptive Design)

3.2 Planned Enrollment & Duration Target \$N\$: The study enrolled 28 participants before it was officially terminated. **Number of Sites:** Multicenter **Estimated Study Start:** 2019 **Estimated Primary Completion:** October 2023 (Terminated early)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female participants who are 18 years old or older at screening. 2. Diagnosed with familial or sporadic ALS according to the World Federation of Neurology Revised El Escorial criteria of either bulbar or limb onset. 3. Disease duration from symptoms onset no longer than 48 months at the screening visit. 4. Females of childbearing potential must have a negative pregnancy test at screening and/or baseline. 5. Treatment with approved ALS therapies is allowed, provided participants have been on a stable dose and regimen for at least 30 days prior to baseline.

4.2 Exclusion Criteria 1. Presence of human immunodeficiency virus (HIV) infection, active or latent tuberculosis, or positive serology for hepatitis B or C. 2. Cardiac disorders, such as recent acute coronary syndrome, heart failure (NYHA class 3 or 4), reduced LVEF, severe pulmonary hypertension, or history of active cardiac valve disorder. 3. Active GI conditions (e.g., Barrett's esophagus, achalasia, esophageal varices, esophageal cancer) or pre-existing pancreatic disease. 4. History of active vasculitis or autoimmune disease associated with vasculitis (e.g., RA, SLE, Sjögren's disease, scleroderma). 5. Pregnant or nursing females, and use of systemic anticoagulation that cannot be temporarily paused before study procedures.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multiple sequential daily doses of BLZ945 (sotuletinib) evaluated in an adaptive cohort design. **Route of Administration:** Oral **Duration of Treatment:** Core Treatment Period, with an optional Extended Treatment Period for eligible patients.

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of approved ALS therapies (e.g., riluzole, edaravone). **Prohibited:** Initiation of a new approved ALS therapy within the next 24 weeks disqualified patients from the Extended Treatment Period. Un-pausable systemic anticoagulation was prohibited.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Up to Day 0 Informed Consent, El Escorial criteria check, Quantiferon testing, viral serologies, LVEF/Cardiac evaluation Baseline Day 0 Baseline TSPO PET scan ([11C]-PBR28), safety labs, first dose of BLZ945 Core Treatment Period Defined interval Oral dosing, ongoing safety assessments, pharmacokinetic sampling (Tmax), follow-up PET scans End of Study / Extended TBD based on cohort Final vital

signs, LVEF evaluation, transition to Extended Treatment Period or study exit |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Characterization of brain microglia response, general safety, and tolerability following multiple doses of the study drug. **Measurement Method:** Positron Emission Tomography (PET) using the TSPO-binding radioligand [11C]-PBR28, alongside clinical safety monitoring and AE tracking.

7.2 Secondary & Exploratory Endpoints - Mean change from baseline in Left Ventricular Ejection Fraction (LVEF). - Pharmacokinetics parameters, specifically Tmax (Time to Reach Maximum Concentration) of BLZ945.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine tracking of clinical parameters, hematological lab abnormalities, signs of liver injury, and cardiac events. **Serious Adverse Events (SAEs):** Continuous evaluation of serious safety signals. (Note: Trial was terminated following a holistic assessment of the potential benefit-risk profile from available safety data). **Data Monitoring Committee (DMC):** Internal and external safety review boards overseen by Novartis Pharmaceuticals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all enrolled subjects who received at least one dose of BLZ945, N=28). **Sample Size Justification:** Adaptive design evaluating successive cohorts (e.g., blocks of four participants) to securely map dose-limiting toxicities and microglia response signals before broader expansion. **Handling of Missing Data:** Standard protocol adaptations for early termination and right-censored data due to the sponsor halting the trial.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring appropriate for Phase II adaptive designs involving radioligands and central nervous system assessments. **Audit Trail:** Captured within Novartis centralized clinical databases, including neuroimaging central reads and eCRF locking.

11. Study Identifiers & Governance

Primary ID: CBLZ945C12201 **Registry Number:** NCT04066244 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** Study of Safety and Proof of the Mechanism of BLZ945 in ALS Patients
Official Regulatory Title: An open-label, adaptive design study in patients with amyotrophic lateral sclerosis (ALS) to characterize safety, tolerability and brain microglia response, as measured by TSP0 binding, following multiple doses of BLZ945, using positron emission tomography (PET) with the radioligand [11C]-PBR28

12. Clinical Framework (ALS Specific)

Primary Condition: Amyotrophic Lateral Sclerosis (ALS) **Diagnosis Threshold:** Familial or sporadic ALS per Revised El Escorial criteria **Medical Methodology:** Microglial target engagement and neuroinflammation modulation **Optimization Target:** Brain microglia response suppression via BLZ945 (sotuletinib)

13. Study Procedures & Methodology

Management Pathway: Stepwise adaptive dose escalation (Core Treatment) followed by maintenance (Extended Treatment) **Education Protocol (HCP):** Site-level training for [11C]-PBR28 PET imaging acquisition and standardization **Education Protocol (Patient):** Informed consent processes emphasizing radioligand properties and compliance with complex visit procedures **Quality Audit Mechanism:** Central review of PET scans and echocardiograms (for LVEF safety monitoring)

14. Observation & Follow-Up Schedule

Total Duration: Limited due to early study termination **Onsite Visit Cadence:** Clustered heavily around Core Treatment Period to facilitate successive daily dosing and immediate PET evaluations **Remote Monitoring Frequency:** Continuous evaluation of the benefit-risk safety profile **Checklist Review Interval:** Routine adverse event and concomitant medication checks at every clinical interaction

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				